



TIME SERIES ANALYSIS

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Fourier Series/Fourier Transform

- **Idea:** Breakdown a signal into weighted combinations of sinusoids with different frequencies:
 - ▶ Similar to expressing $x[n]$ as a combination of impulses
 - ▶ In FT, the basis functions are sinusoids instead of impulses
- **Significance**
 - ▶ **Weights or coefficients are in general complex**
 - ▶ Magnitude of weights give the energy or power of that frequency component in the signal - **Spectral Analysis**
 - ▶ Angle of coefficients give how much each sinusoid (at that frequency) in the signal is aligned with respect to the basis sinusoids. Useful in time-delay or lag estimation.





From a Correlation perspective

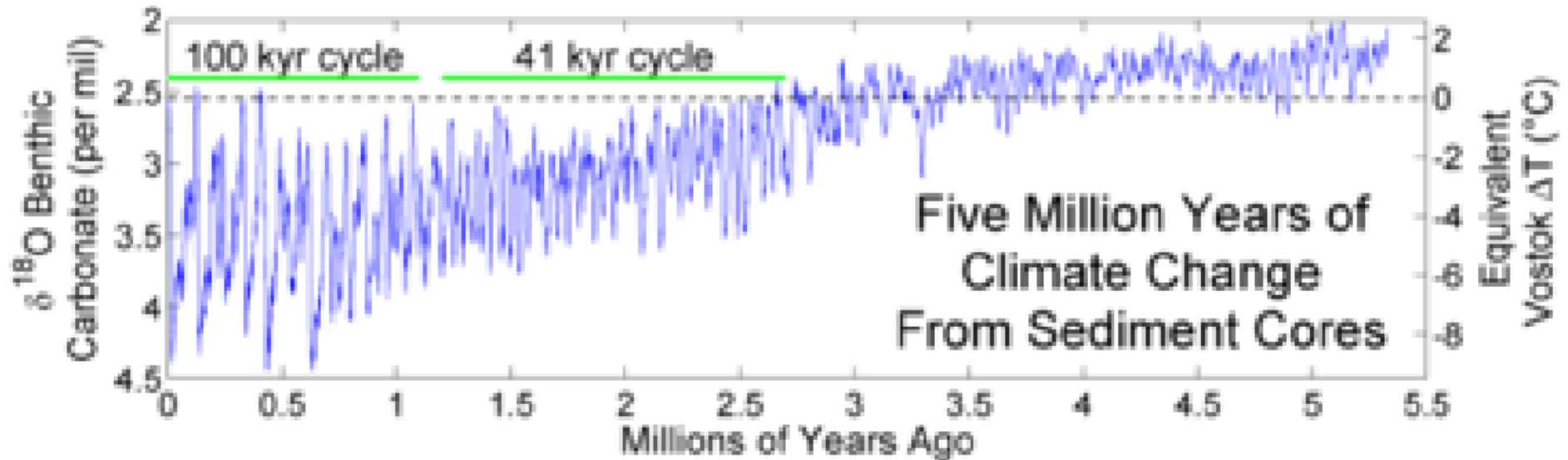
Every transform of a signal $x[n]$ can be viewed as a correlation of that signal $x[n]$ with the basis function. The coefficient of transform is the “amount” of correlation or similarity of that signal with the basis function

- The breaking down of a signal is equivalent to finding the best set of sinusoids that can explain the pattern in that signal.
- In Fourier transform, each basis function is a (complex) sinusoid of a certain frequency.
- The coefficient at each frequency is a measure of similarity between the signal and the sinusoid
- The complex sinusoid is chosen so that similarity in shift and magnitude both are taken into account
 - ▶ The sine and cosine pair form an orthogonal set of basis functions



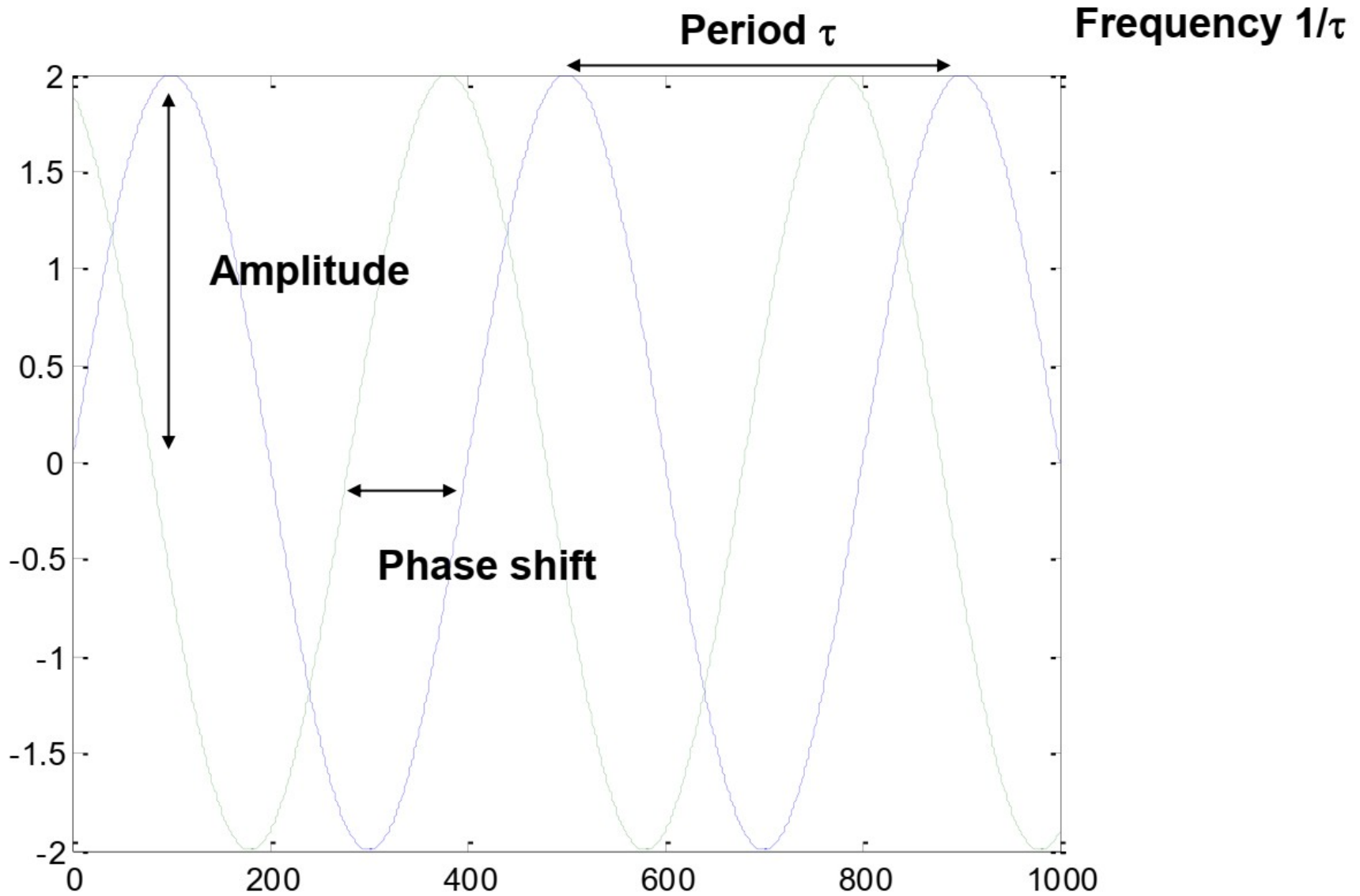
Frequency domain analysis

- Time domain: Straight forward, data over time
- Frequency domain: Data decomposed to sine/cos waves.





Periodicity





Continuous-time Periodic Signals

- **idea:** A continuous-time periodic signal is expressed as a (linear weighted combination of harmonics:

$$x(t) = \sum_{k=-\infty}^{\infty} c_k \exp(j2\pi k F_0 t) \quad (\text{Fourier Series})$$

- F_0 : Fundamental frequency of $x(t)$
- Calculation of c_k :

$$c_k = \frac{1}{T_p} \int_{T_p} x(t) e^{-j2\pi k F_0 t} dt$$

- c_k is in general a complex quantity.
- In order to perform the above calculation, an analytical expression for $x(t)$ needs to be provided



Why do negative frequencies come in?

- In the expression for $x[n]$ earlier, $x(t) = \sum_{k=-\infty}^{\infty} c_k \exp(j2\pi k F_0 t)$ we notice that the summation includes both negative and positive frequencies
- The need for including both negative frequencies can be easily seen by writing the expression for a sinusoid:

$$\sin(2\pi F_0 t) = \frac{1}{2j} (e^{j2\pi F_0 t} - e^{-j2\pi F_0 t})$$

Observe that two exponentials, one with a positive frequency F_0 and the other with a negative frequency $-F_0$ are required to explain a sinusoid

- The corresponding coefficients are c_1 ($k = 1$) and c_{-1} are $\frac{1}{2j}$ and $-\frac{1}{2j}$
- Therefore, in general, Fourier series / transform involves expressing any signal as **addition** and **subtraction** of cosines and sines



Important Quantities

- A periodic signal has infinite energy, but finite power given by

$$P_{xx} = \frac{1}{T_p} \int_0^{T_p} x^2(t) dt$$

- Fourier transform allows one to split this overall power into power contributions by different frequencies (harmonics)
- **Power spectrum:** The average power satisfies

$$P_{xx} = \frac{1}{T_p} \int_0^{T_p} x^2(t) dt = \sum_{k=-\infty}^{\infty} |c_k|^2 \quad (\text{Parseval's relation})$$

- $P_k = |c_k|^2$ is the contribution by the k^{th} harmonic and hence known as the power spectral density or simply *power spectrum*
- **Phase:** $\theta_k = \angle c_k$
 - ▶ A plot of θ_k vs. k shows how each frequency component is aligned w.r.t the basis functions
- As $T_p \rightarrow \infty$, $x(t)$ becomes an *aperiodic* signal and frequency spacing tends to zero



Power Spectrum

- The power in a periodic signal exists only at discrete frequencies.
Hence the power spectrum is also known as *line spectrum*
- $c_k = |c_k|e^{j\theta_k}$
- c_0 therefore represents the average component of the signal
- The power spectral density plot is independent of (or blind to) the phase
 - ▶ *Two signals having two different phases but same strengths will have identical power spectral densities*
- For a real-valued signal, $c_k^* = c_{-k}$
 - ▶ Power spectrum of any measurement will therefore be symmetric with respect to frequency
- If the periodic function is even, the coefficients c_k are real



Continuous-time Aperiodic Signal

- **idea:** A continuous-time aperiodic signal with finite duration is expressed as a (linear) weighted combination of harmonics:

$$x(t) = \int_{-\infty}^{\infty} X(F) e^{j2\pi Ft} dF$$

- $X(F)$ is similar to c_k in Fourier Series
- Calculation of $X(F)$ (integrate over entire time interval)

$$X(F) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi Ft} dt \quad (\text{Fourier Transform})$$

- Aperiodic signals are Energy signals
 - ▶ Plot of $S_{xx} = |X(F)|^2$ vs. F is known as the **energy density plot**





Conditions for Existence of Fourier Transform

A set of conditions have to be met to guarantee the existence of Fourier transform (known as the *Dirichlet's conditions*)

- ① The signal $x(t)$ has a finite number of finite discontinuities
- ② The signal $x(t)$ has a finite number of maxima and minima
- ③ The signal $x(t)$ is absolutely integrable, that is,

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

- The above conditions are sufficient, but not necessary, for the existence of Fourier transform
- Nearly all finite energy signals have a Fourier transform



Frequency domain analysis of discrete-time signals

The Fourier series representation for discrete-time signals has some similarities with that of continuous-time signals. Nevertheless, certain differences exist:

- Discrete-time signals are unique over the frequency range $(-\pi, \pi)$ or $(0, 2\pi)$
- The period of a discrete-time signal can only be expressed in samples rather than time units
- A discrete-time signal of fundamental period N can consist of frequency components separated by $2\pi/N$ radians or $f = 1/N$ cycles
 - ▶ Therefore, the Fourier series representation of the discrete-time periodic signal will contain at most N frequency components



Frequency domain analysis of discrete-time signals

Given a periodic sequence $x[n]$ with period N , the Fourier series representation for $x[n]$ consists of N harmonically related exponential functions

$$e^{j2\pi kn/N} \quad k = 0, 1, \dots, N-1$$

The Fourier series is expressed as

$$x[n] = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N}$$

- The coefficients $\{c_k\}$ are given by:

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$



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Power Spectral Density

The *average power* of a discrete-time periodic signal with period N is defined as

$$P_{xx} = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2 = \sum_{k=0}^{N-1} |c_k|^2$$

- The term $|c_k|^2$ denotes therefore the power associated with the k^{th} frequency component and hence is known as the **power spectrum density** or *power density spectrum* of $x[n]$
- The energy of $x[n]$ over one period can be computed as
$$E_{xx} = \sum_{n=0}^{N-1} |x[n]|^2 = N \sum_{k=0}^{N-1} |c_k|^2$$
- The above definition is similar to that for the continuous-time periodic signal.



Remarks

- The expression for $x[n]$ is known as the **synthesis equation**, while the expression for c_k is known as the **analysis equation**
- The Fourier series is known as the *Discrete-Time* Fourier Series (DTFS)
- As in continuous-time case, c_k is complex and carries similar connotations
- **Note:** The Fourier coefficients $\{c_k\}$ are periodic with the same period as $x[n]$
 - ▶ *The power spectrum of a discrete-time periodic signal is also, therefore, periodic*
- The range $0 \leq k \leq N - 1$ amounts to the fundamental frequency range $0 \leq \omega_k = 2\pi k/N \leq 2\pi$
 - ▶ The frequency range $-\pi \leq \omega_k = 2\pi k/N \leq \pi$ corresponds to $-N/2 \leq k \leq N/2$ (inconvenient when N is odd)
 - ▶ Also, $0 \leq k \leq N - 1$ corresponds to $0 \leq F \leq F_s$



Fourier Transform of Discrete-time Aperiodic Signals

The Discrete-Time Fourier Transform (DTFT) of a **finite-energy** discrete-time aperiodic signal is given by

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega N} \quad (\text{DTFT})$$

Remarks

- Two basic differences exist between the FT of a finite-energy discrete-time signal and the FT of a finite-energy analog signal
 - ① For analog signals, the FT, and hence the spectrum of the signal have a frequency range of $(-\infty, \infty)$, whereas for the discrete-time signal the FT is unique only in the interval $(0, 2\pi)$ (or $(-\pi, \pi)$)
 - ② The integral in the continuous-time case is replaced with a summation in the discrete-time signal due to the nature of $x[n]$



Fourier Transform of Discrete-time Aperiodic Signals

Periodicity of $X(\omega)$

The DTFT $X(\omega)$ is periodic with period 2π (irrespective of whether the discrete-time sequence $x[n]$ is periodic or not)

- The power (energy density) spectrum of discrete-time signal is periodic with period 2π
- Convergence is guaranteed provided the signal $x[n]$ is absolutely summable (finite 1-norm) or is of finite energy (finite 2-norm, weaker condition)
- The inverse DTFT is given by

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X[\omega] e^{j\omega n} d\omega \quad (\text{Inverse DTFT})$$



Energy Spectrum Density

- The energy of a discrete-time signal $x[n]$ is defined as

$$E_{xx} = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

- This can be shown equal to

$$E_{xx} = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega$$

- The quantity $S_{xx}(\omega) = |X(\omega)|^2$ represents the distribution of energy as a function of frequency, and is termed as the *energy density spectrum* of $x[n]$.
- Since any measurement is real, $X^*(\omega) = X(-\omega)$; therefore, **the energy density spectrum is also symmetric**



Summary

- Fourier transform is identical to evaluating the z -transform of a signal on the *unit circle* $z = e^{j\omega}$
- Continuous-time signals have aperiodic spectra
- Discrete-time signals have periodic spectra
- Periodic signals have discrete spectra
- Aperiodic finite energy signals have continuous spectra



Properties of Fourier Transform

In general, for a *complex-valued* signal $x[n]$, the following hold

$$\begin{aligned} X_R(\omega) &= \sum_{-\infty}^{\infty} [x_R[n] \cos(\omega n) + x_I[n] \sin(\omega n)] \\ X_I(\omega) &= \sum_{-\infty}^{\infty} [x_R[n] \sin(\omega n) - x_I[n] \cos(\omega n)] \end{aligned}$$

$$\begin{aligned} x_R[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} [X_R(\omega) \cos(\omega n) - X_I(\omega) \sin(\omega n)] d\omega \\ x_I[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} [X_R(\omega) \sin(\omega n) + X_I(\omega) \cos(\omega n)] d\omega \end{aligned}$$

- Real-valued even signals possess real-valued Fourier Transforms and are even functions of frequency
- Real-valued odd signals possess purely imaginary-valued Fourier transforms and are odd functions of the frequency



Properties of Fourier Transform

- **Linearity:** If $x_1[n] \xrightarrow{\mathcal{F}} X_1(\omega)$ and $x_2[n] \xrightarrow{\mathcal{F}} X_2(\omega)$ then $a_1x_1[n] + a_2x_2[n] \xrightarrow{\mathcal{F}} a_1X_1(\omega) + a_2X_2(\omega)$

Determine the Fourier transform of $x[n] = a^{|n|}$ where $-1 < a < 1$

- **Time shifting:** If $x_1[n] \xrightarrow{\mathcal{F}} X_1(\omega)$ then $x_1[n - k] \xrightarrow{\mathcal{F}} e^{-j\omega k} X_1(\omega)$
 - ▶ Note that the power spectrum of the shifted signal remains unchanged while the phase spectrum shifts at each frequency by $-\omega k$
- **Time reversal:** If $x[n] \xrightarrow{\mathcal{F}} X(\omega)$, then $x[-n] \xrightarrow{\mathcal{F}} X(-\omega)$
 - ▶ If a signal is folded in time, then its power spectrum remains unchanged; however, the phase spectrum undergoes a sign reversal



Convolution Theorem

- **Convolution Theorem:** If $x_1[n] \xrightarrow{\mathcal{F}} X_1(\omega)$ and $x_2[n] \xrightarrow{\mathcal{F}} X_2(\omega)$ then the Fourier Transform of $x[n] = x_1[n] \star x_2[n] = \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k]$ is given by

$$x[n] = x_1[n] \star x_2[n] \xrightarrow{\mathcal{F}} X_1(\omega)X_2(\omega)$$

- Convolution in time-domain transforms into a product in the frequency domain



Correlation Theorem

Wiener Khintchine Theorem

Fourier Transform of auto-covariance function (ACF) of a sequence $x[n]$ is the energy density spectrum (power spectral density) of that sequence

If $\sigma_{xx}[l]$ is the ACF of the sequence $x[n]$ then $\sigma_{xx}[l] \xrightarrow{\mathcal{F}} S_{xx}(\omega)$

Sample problems

- 1 Determine the energy density spectrum of the signal $x[n] = a^n u[n]$ where $-1 < a < 1$
- 2 By use of the above property, determine the power spectrum of a moving average random process $v[k] = e[k] + c_1 e[k-1]$ where $e[k]$ is the *white-noise* sequence



Cross Spectral Density

Wiener Khintchine Theorem

The Fourier Transform of cross-correlation function (CCF) of two sequences $x_1[n]$ and $x_2[n]$ is the cross energy density (power spectral density) spectrum of those sequences

If $\sigma_{x_1x_2}[l]$ is the CCF of the sequence $x[n]$ then $\sigma_{x_1x_2}[l] \xrightarrow{\mathcal{F}} S_{x_1x_2}(\omega)$

- The auto (cross)-correlation sequence of a signal and the energy density (power spectral density) contain the same information



Frequency Shifting and Modulation

- **Frequency shifting:** If $x[n] \xrightarrow{\mathcal{F}} X(\omega)$ then $e^{j\omega_0 n} x[n] \xrightarrow{\mathcal{F}} X(\omega - \omega_0)$

- **Modulation:** If $x[n] \xrightarrow{\mathcal{F}} X(\omega)$ then
 $x[n] \cos(\omega_0 n) \xrightarrow{\mathcal{F}} \frac{1}{2}[X(\omega + \omega_0) + X(\omega - \omega_0)]$

- **Multiplication of two sequences:**

$$x[n] = x_1[n]x_2[n] \xrightarrow{\mathcal{F}} \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(\lambda)X_2(\omega - \lambda) d\lambda$$

- **Differentiation in frequency domain:**

$$\text{If } x[n] \xrightarrow{\mathcal{F}} X(\omega) \text{ then } nx[n] \xrightarrow{\mathcal{F}} j \frac{dX(\omega)}{d\omega}$$



Parseval's Theorem

- **Parseval's identity:** If $x_1[n] \xrightarrow{\mathcal{F}} X_1(\omega)$ and $x_2[n] \xrightarrow{\mathcal{F}} X_2(\omega)$ then

$$\sum_{n=-\infty}^{\infty} x_1[n]x_2^*[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(\omega)X_2^*(\omega) d\omega$$

An important observation results from the above,

$$E_{xx} = \sigma_{xx}[0] = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{xx}(\omega) d\omega$$

- The variance of a signal is equal to the area under the energy density curve



Time-Frequency Relationship

- Every waveform in the time-domain has a counterpart in the frequency-domain
 - ▶ For example, a finite duration rectangular pulse in the time-domain has a *sinc* function $\frac{\sin x}{x}$ as its counterpart in the frequency domain
- It is useful to remember such pairs

Time-domain	Frequency-domain
Delta function $\delta(t)$	Constant <i>spectrum</i> with zero phase
Shifted impulse $\delta(t - \tau)$	Constant spectrum with linear phase
Rectangular pulse	<i>Sinc</i> function
Sinc function $\frac{\sin(2\pi f_c k)}{\pi k}$	Rectangular pulse with cut-off frequency f_c

Time-Frequency Relationship

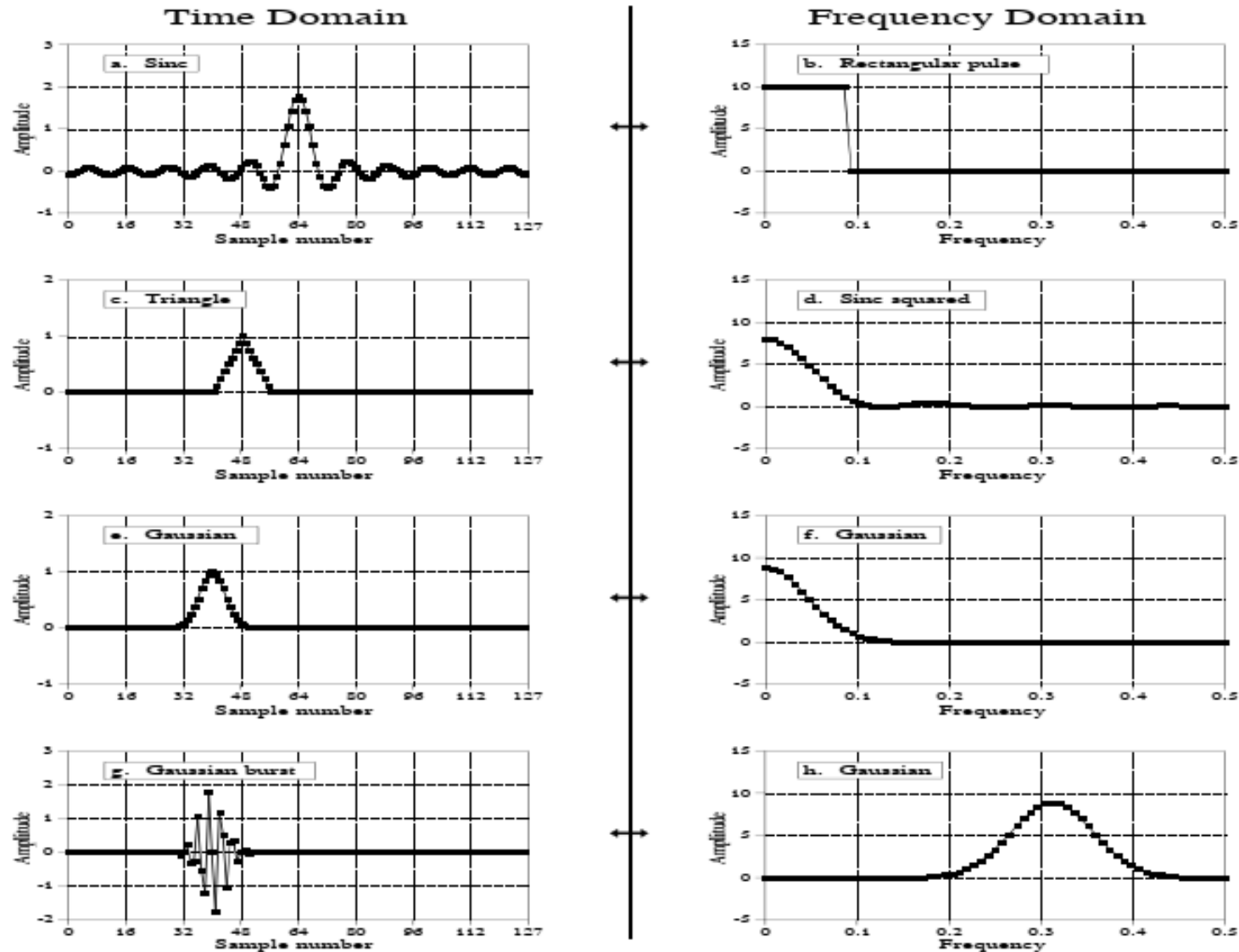


FIGURE 11-5



Gibbs Relation

- Gibbs effect is concerned with the reconstruction of the signal from its Fourier transform
- Of particular interest are the edges when they are reconstructed with fewer frequencies than used in the transform
- When only some frequencies are chosen for reconstruction, the edges show *overshoot* and ringing
 - ▶ This overshoot and ringing is jointly termed as the **Gibbs effect**
 - ▶ For discrete-time signals, the *width* of this overshoot decreases with addition of more frequencies while the *amplitude* vanishes only when the last frequency is added
 - ▶ For continuous-time signals, the *amplitude* of oscillations never dies out with the addition of even infinite sinusoids, but the *width* dies out
- Gibbs effect has implications in the filter domain and also addresses the question:
 - ▶ Can an edge be reconstructed using summation of sinusoids?

Gibbs Relation

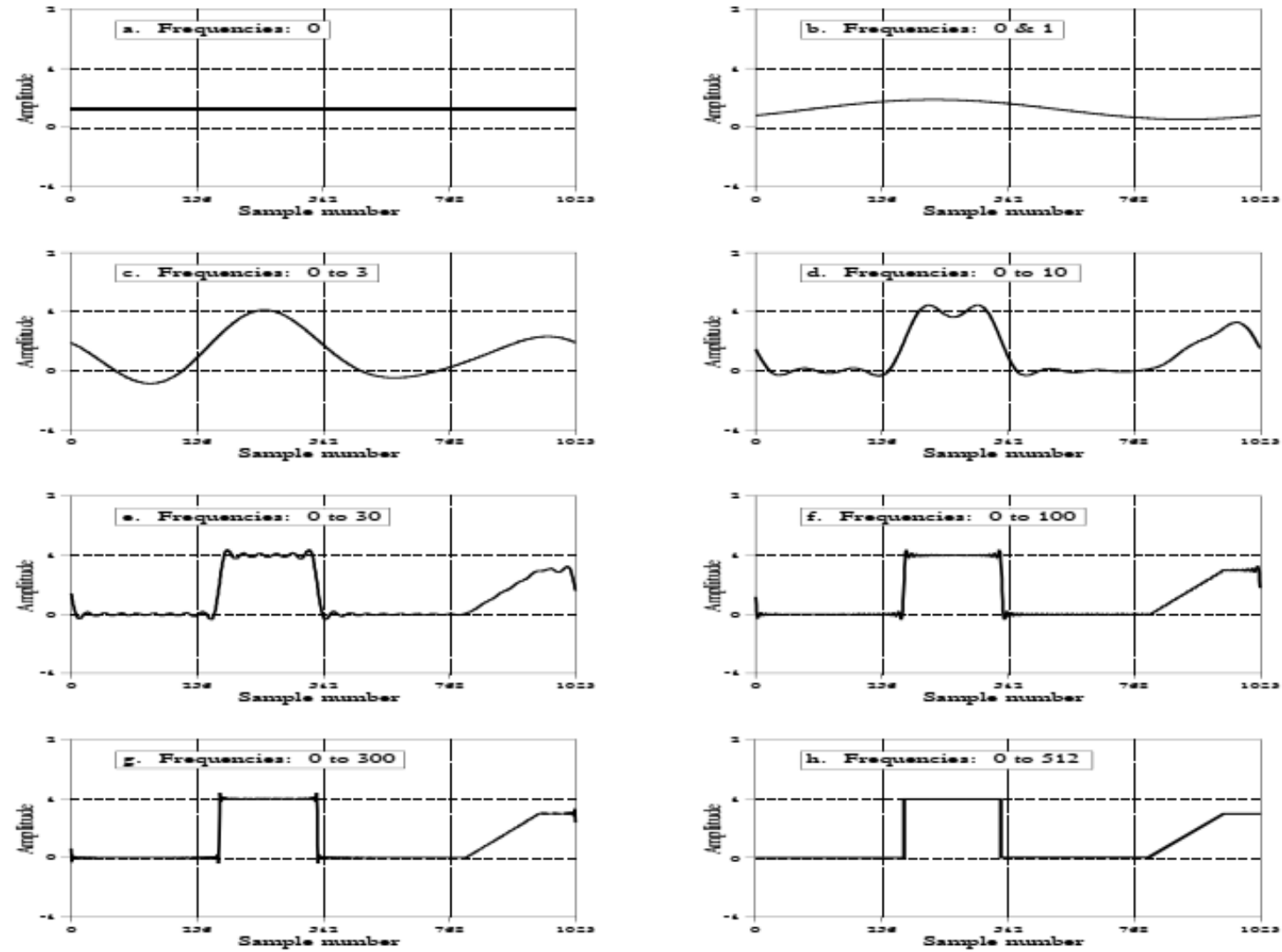


FIGURE 11-6.
The Gibbs effect.



Harmonics

- Fourier transform always expands (or expresses) a given signal in terms of a *linear* combination of its building blocks which are sinusoids
- What would you expect to see in the power spectrum for signals such as: (i) $\cos^2(2\pi(0.02)n)$, (ii) $\cos^2(2\pi(0.2)n)$, (iii) $\cos^2(2\pi(0.3)n)$ and (iv) $\cos^3(2\pi(0.3)n)$
- Similarly, what happens when the time-domain signal contains *harmonics* much greater than the fundamental frequency?
- In some of the situations above, aliasing is typically noticed (depending on the frequencies)
 - ▶ For example, in case of $\cos^2(2\pi(0.3)n)$, the linear expansion contains $\cos(2\pi(0.6)n)$ which appears as $\cos(2\pi(0.4)n)$!



Discrete Fourier Transform (DFT)



Auto Spectral Analysis: Discrete Fourier transform (DFT)

- Discrete values over a limited duration (scientific data)
- Fast Fourier transform (FFT) algorithm is usually used for DFT in practice
- Matlab
 - $Y = \text{fft}(X)$
 - X: time domain
 - Y: frequency domain





Recall...

- Recall that while periodic (discrete-time) signals have *discrete spectra*, aperiodic finite energy signals have *continuous spectra*, i.e., the frequency axis is continuous

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

- However, it is not practical to compute the above quantity for all points along continuous frequency axis
- Some form of discretization of the frequency axis, i.e., *sampling in frequency* is therefore inevitable.
- In practice, since $X(\omega)$ is periodic with period 2π , it is sufficient to sample the frequency axis in the fundamental range.
- The resulting sampled DTFT is known as **Discrete Fourier Transform (DFT)**
 - ▶ Note that the transform is now discrete in both time and frequency



DFT

- If the DTFT is evaluated at N equidistant points (spacing $\delta\omega = \frac{2\pi}{N}$ along the frequency axis, one obtains

$$\begin{aligned} X\left(\frac{2\pi}{N}k\right) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j2\pi kn/N} \quad k = 0, 1, \dots, N-1 \\ &= \sum_{l=-\infty}^{\infty} \sum_{n=lN}^{lN+N-1} x[n]e^{-j2\pi kn/N} \\ &= \sum_{n=0}^{N-1} \sum_{l=-\infty}^{\infty} x[n-lN]e^{-j2\pi kn/N} \end{aligned}$$

- The signal $x_p[n] = \sum_{l=-\infty}^{\infty} x[n-lN]$ is clearly periodic with period N
- Therefore, a Fourier series expansion can be written for $x_p[n]$

$$x_p[n] = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N} \quad c_k = \frac{1}{N} \sum_{n=0}^{N-1} x_p[n] e^{-j2\pi kn/N} \quad k = 0, 1, \dots, N-1$$



DFT

Upon comparison of equations, we obtain $c_k = \frac{1}{N} \sum_{n=0}^{N-1} x_p[n] e^{-j2\pi kn/N}$

Basic Idea

- 1 Create a periodic signal $x_p[n]$ of period N from the given finite-duration signal of length L
 - 2 Compute the Fourier series coefficients of the resulting periodic signal
- The resulting coefficients are the values of the DFT and the above method is known as the N -point DFT
 - It is important to ensure that $N \geq L$ to avoid any loss of information of the given time-domain sequence $x[n]$



Reconstruction

- Once the discretized DTFT is obtained, it is possible to reconstruct the continuous spectrum
- The reconstruction formula is given by

$$X(\omega) = \sum_{k=0}^{N-1} X\left(\frac{2\pi}{N}k\right) P\left(\omega - \frac{2\pi}{N}k\right) \quad N \geq L$$

$$\text{where } P(\omega) = \frac{\sin(\omega N/2)}{N \sin(\omega/2)} e^{-j\omega(N-1)/2}$$

- The above reconstruction is exactly identical to the way a continuous-time signal $x(t)$ is reconstructed from its samples $x[n]$
- The condition $N \geq L$ is similar to the aliasing condition (in frequency domain) when sampling a continuous-time signal



Summary

Definition

The N-point DFT and IDFT are given by

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$
$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}$$

where it is assumed that $x[n]$ is padded with $(N - L)$ zeros to bring it up to length N

- The above relationships are also sometimes written as

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{-kn} \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{kn}$$

where $W_N = e^{j2\pi/N}$



Frequency domain analysis of discrete-time signals

$$\text{DFT} \quad f(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos \frac{2n\pi}{T} t + b_n \sin \frac{2n\pi}{T} t \right]$$

Forward

$$X = \text{fft}(x)$$

$$X(k) = \sum_{j=1}^N x(j) \omega_N^{(j-1)(k-1)}$$

Backward

$$x = \text{ifft}(X)$$

$$x(j) = \frac{1}{N} \sum_{k=1}^N X(k) \omega_N^{-(j-1)(k-1)}$$

where

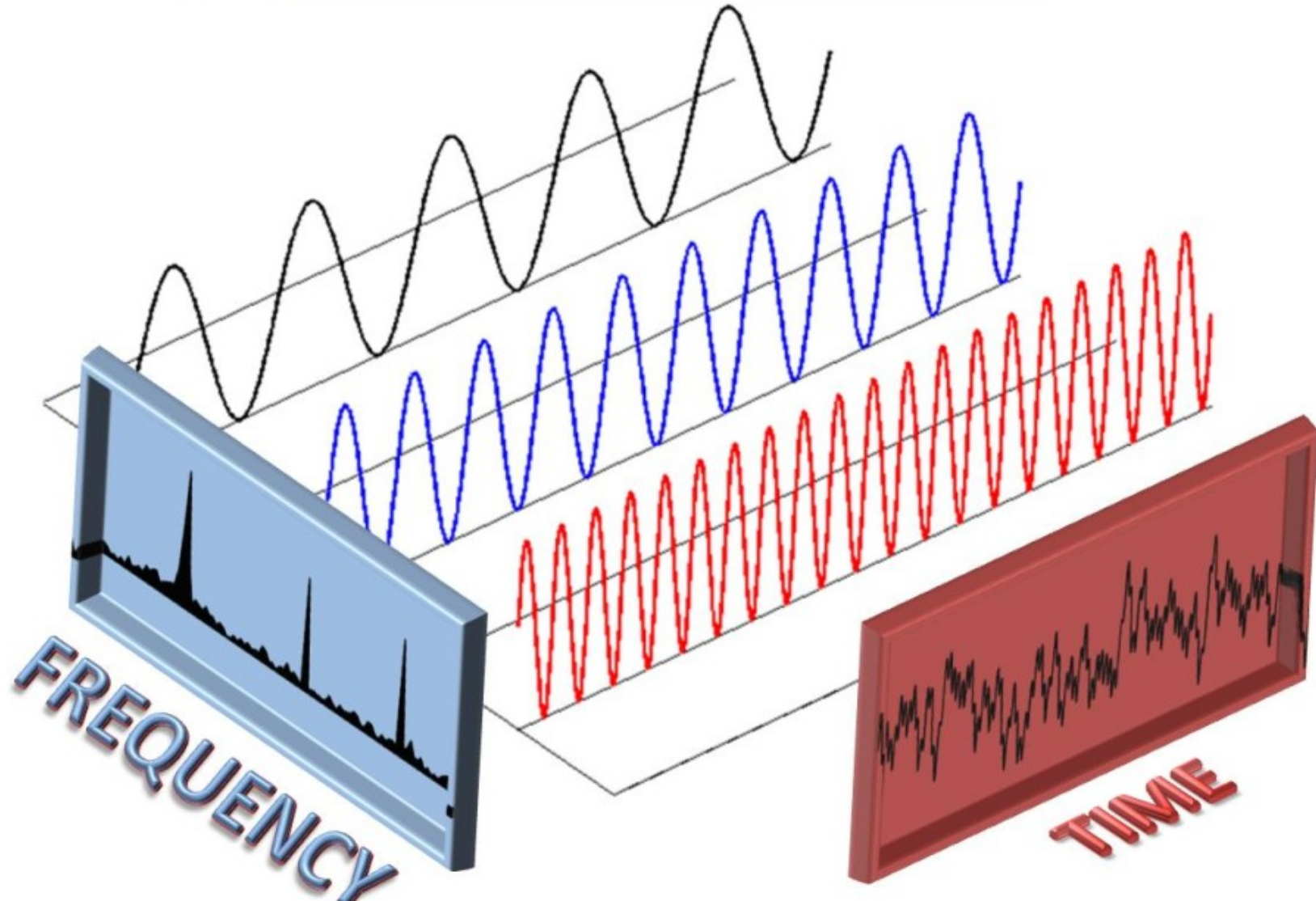
$$\omega_N = e^{(-2\pi i)/N}$$

$$e^{i\theta} = \cos(\theta) + i \sin(\theta)$$

$$F = \begin{bmatrix} \omega_N^{0 \cdot 0} & \omega_N^{0 \cdot 1} & \dots & \omega_N^{0 \cdot (N-1)} \\ \omega_N^{1 \cdot 0} & \omega_N^{1 \cdot 1} & \dots & \omega_N^{1 \cdot (N-1)} \\ \vdots & \vdots & \ddots & \vdots \\ \omega_N^{(N-1) \cdot 0} & \omega_N^{(N-1) \cdot 1} & \dots & \omega_N^{(N-1) \cdot (N-1)} \end{bmatrix}$$

$$X = Fx; \quad x = F^{-1}X = \frac{1}{N}F^*X$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos \frac{2n\pi}{T} t + b_n \sin \frac{2n\pi}{T} t \right]$$



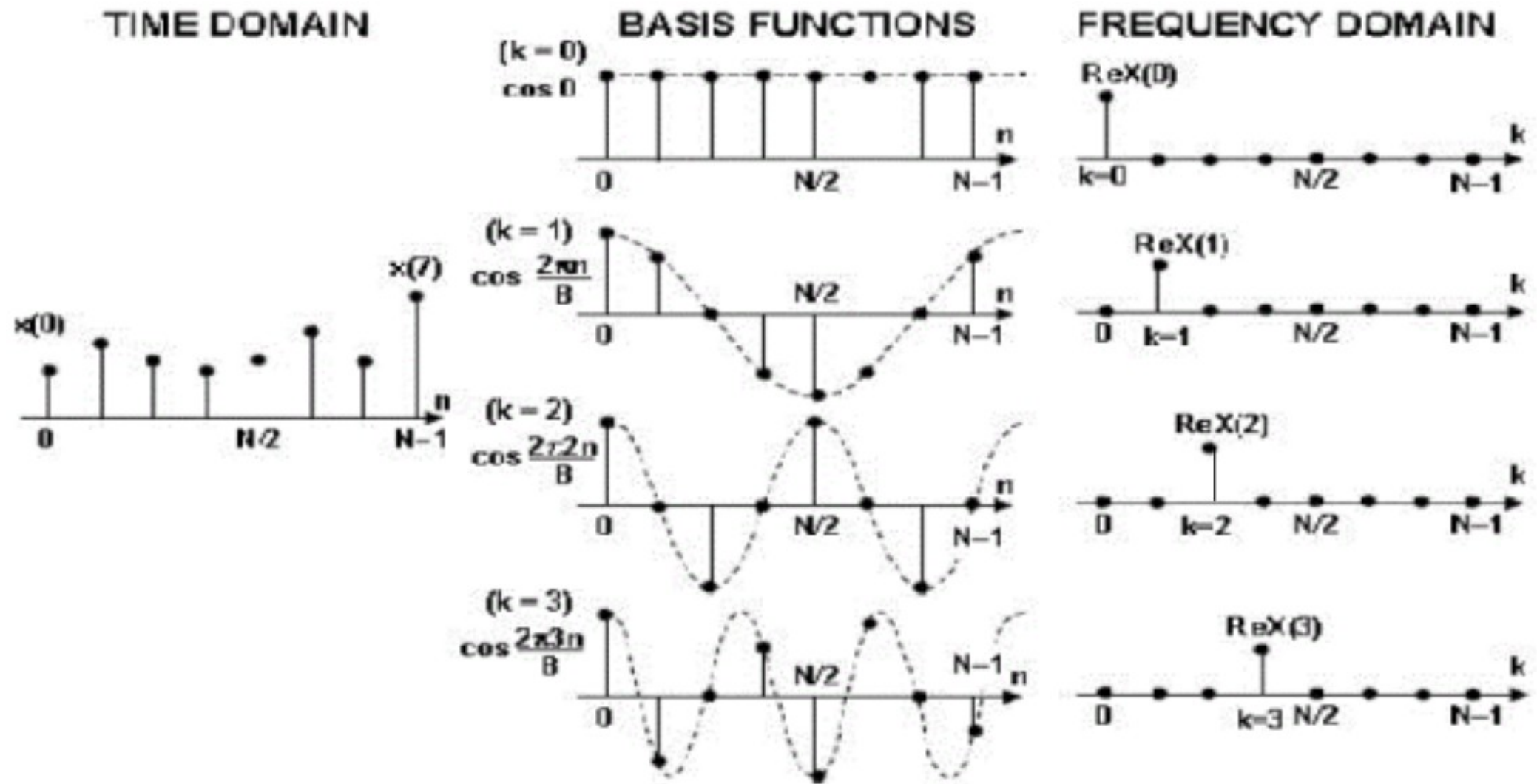
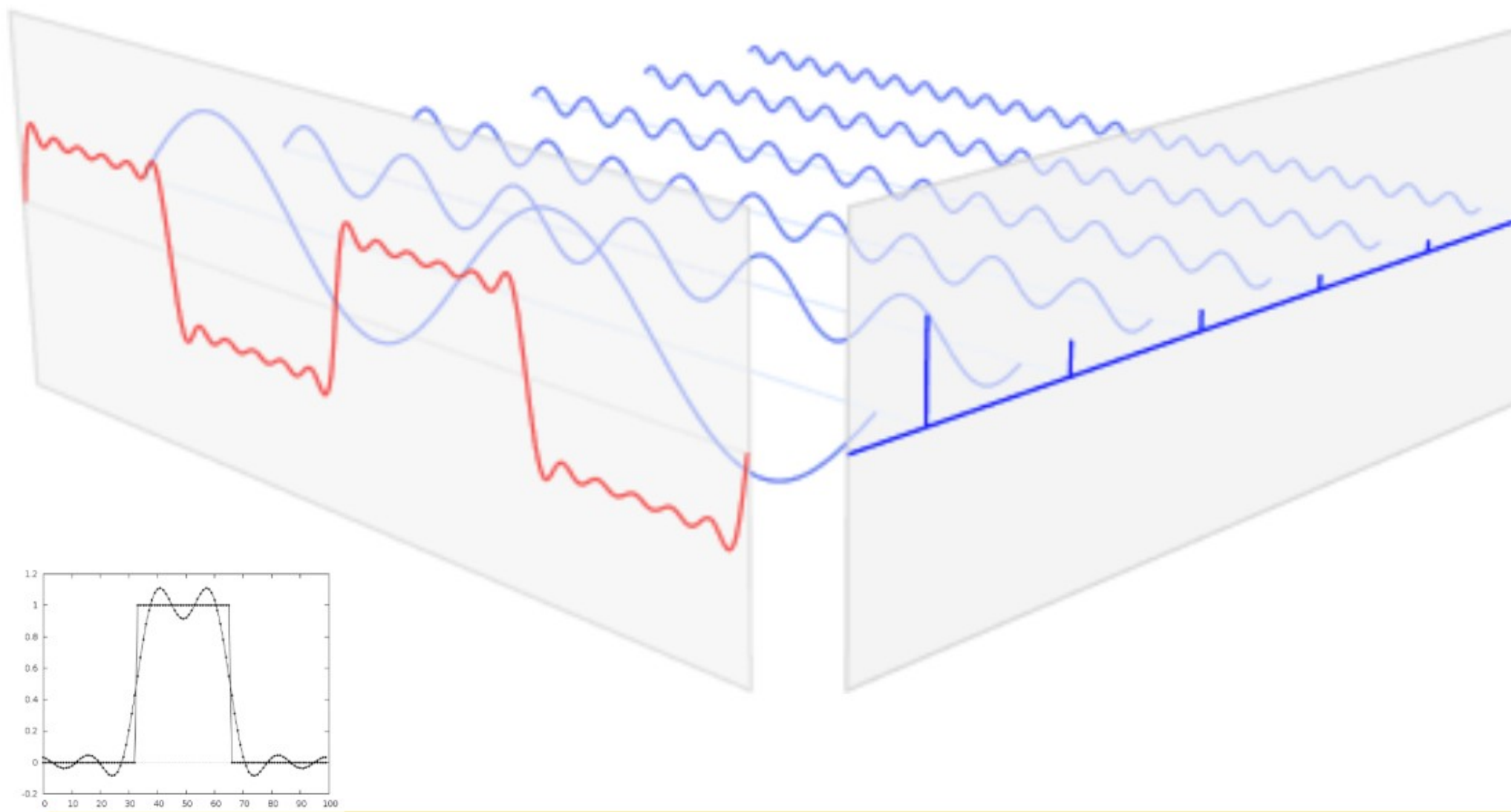


Figure 03: Correlation of time samples with basic functions using the DFT





FFT

The time interval of data collection is τ and we have a total of N data. The number of frequencies that can be resolved by FFT ($fx=fft(x,[])$) is

$$\text{Traditional } nfft = 2^{\text{ceil}(\log_2 N)}$$

$N = 1000$, $nfft=1024$; If $N=200$, what is $nfft$?

We set $nfft=n$



FFT

Nyquist frequency: the highest frequency resolved by FFT is $1/2f_s$, where f_s is sampling frequency.

The lowest frequency that can be resolved is $1/\tau$, where τ is sampling period

If data are sampled at every 5 seconds, what is the Nyquist frequency f_N ?



FFT

The highest frequency resolved by FFT is

$$f_{\max} = (1/2 - 1/n)f_s < f_N$$

The frequencies resolved by FFT is

$$f = f_s * \text{linspace}(0, 1/2 - 1/n, n/2)$$

$$\text{period} = 1./f$$

How to design sampling strategy?

Compare $f_s=0.2$ & $n=10$ with $f_s=0.4$ & $n=10$

Compare $f_s=0.4$ & $n=10$ with $f_s=0.4$ & $n=20$



FFT

- $f_s=0.2$ & $n=10$ (different sampling frequencies)

$f = 0 \quad 0.0200 \quad 0.0400 \quad 0.0600 \quad 0.0800$

- $f_s=0.4$ & $n=10$

$f = 0 \quad 0.0400 \quad 0.0800 \quad 0.1200 \quad 0.1600$

- $f_s=0.4$ & $n=10$ (different sampling periods)

$f = 0 \quad 0.0400 \quad 0.0800 \quad 0.1200 \quad 0.1600$

period = Inf 25 12.5 8.33 6.25 (s)

- $f_s=0.4$ & $n=20$

$f = 0 \quad 0.0200 \quad 0.0400 \quad 0.0600 \quad 0.0800 \quad 0.1000 \quad 0.1200 \quad 0.1400$
 $0.1600 \quad 0.1800$

period = Inf 50 25 16.7 12.5 10 8.33 7.14 6.25 5.56(s)

- Higher $f_s \rightarrow$ higher f
- Higher $n \rightarrow$ higher f resolution



FFT

We generate a sine function of 1000 data points (sampled at 1 Hz)

```
N = 1000;
```

```
t = 0 : N-1;
```

```
y = sin(2*pi*t/100);
```

```
plot(t,y)
```

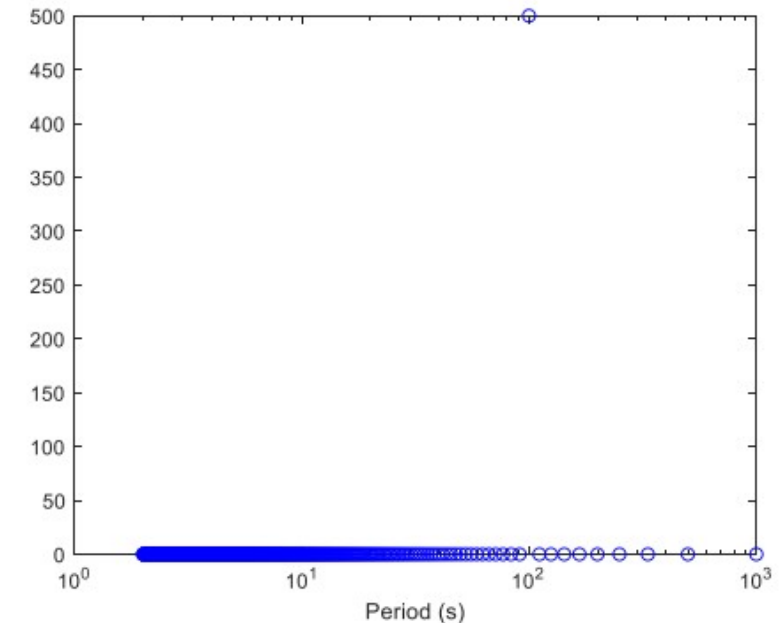
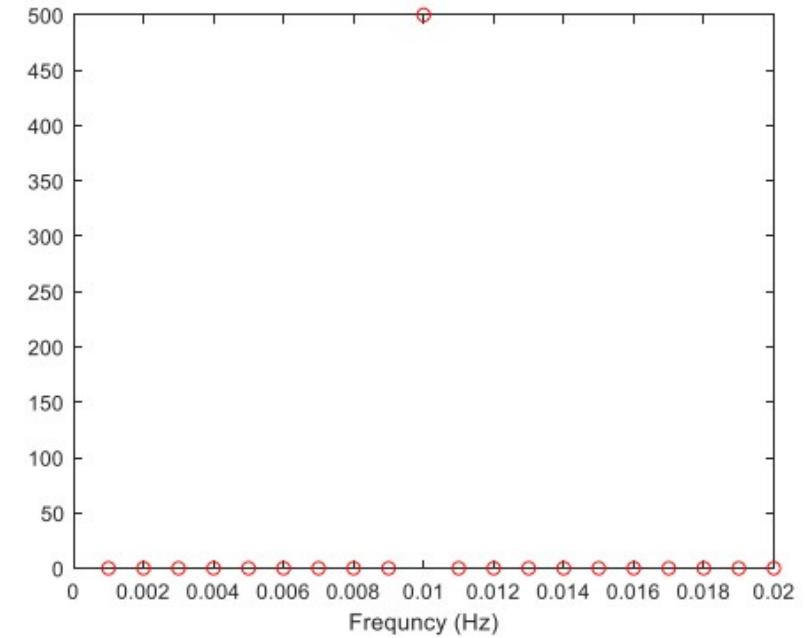
Use: $f_y = \text{fft}(y, n_{\text{fft}})$

Plot **abs**(f_y) as a function of frequency

Why do we not want to plot $f_y(1)$?



```
nfft=N;  
fy = fft(y,nfft);  
fs = 1; %1 sec per data  
f=fs*linspace(0,1/2-1/nfft,nfft/2);  
plot(f(2:21),abs(fy(2:21)),'ro')  
xlabel('Frequency (Hz)')  
figure  
nn=2:nfft/2; %fy(1); is constant  
semilogx(1./f(nn),abs(fy(nn)),'bo')  
xlabel('Period (s)')
```





DFT as linear transformation

- It is useful to view DFT and IDFT as linear transformations

$$\mathbf{x}_N = \mathbf{W}_N \mathbf{X}_N \quad \mathbf{X}_N = \frac{1}{N} \mathbf{W}_N^* \mathbf{x}_N$$

where the quantities $\mathbf{x}_N$, $\mathbf{X}_N$ and $\mathbf{W}_N$ are stacked up as

$$\mathbf{x}_N = \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}; \quad \mathbf{X}_N = \begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix}$$
$$\mathbf{W}_N = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & W_N & W_N^2 & \dots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & \dots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \dots & W_N^{(N-1)(N-1)} \end{bmatrix}$$



Points to Remember

- The frequency resolution in DFT is equal to $1/N$ or $2\pi/N$. Padding with zeros does not provide any new information but provides a better “display” of the spectrum
- DFT is calculated assuming that the given signal $x[n]$ is periodic and therefore it is a *Fourier series* expansion of $x[n]$ in reality!
- In an N -point DFT, only $N/2 + 1$ frequencies are unique. For example, in a 1024-point DFT, only 513 frequencies are sufficient to reconstruct the original signal.
 - ▶ This is due to the symmetricity of the Fourier transform
- The basis blocks are $\cos(2\pi \frac{k}{N} n)$ and $\sin(2\pi \frac{k}{N} n)$ characterized by the parameter k
 - ▶ The quantity k denotes the number of cycles completed by each basis block for the duration of N samples
 - ▶ The highest value for k (after which repetition occurs) is $N/2$
 - ▶ The value at $k = 0$ always corresponds to the DC component



DFT in Practice

- The linear transformation relationships are useful for short calculations
- Until late '60s, there were no sophisticated algorithms to implement these calculations
- In 1960s, Cooley and Tukey developed an efficient algorithm for fast computation of DFT which revolutionized the world of spectral analysis
 - ▶ This algorithm and its subsequent variations came to be known as the Fast Fourier Transform (FFT), which is available today in every good numerical computation package
- The FFT algorithm reduced the number of operations from N^2 in regular DFT to the order of $N \log(N)$
- FFT is merely an algorithm, but not a new transform!
- FFT algorithms are fast when N is exactly a power of 2
 - ▶ Therefore, in practice, the value of N is chosen as the next higher power of 2 closest to L